

Part I

Introduction

This work is a thesis on a new system OPEN-RMF still in development
applied for a robot in the facility Elettra Sincrotrone Trieste

Chapter 1

Background and motivation

Elettra Sincontrone Trieste¹ is a big research facility that, principally, gives to users two light sources: the synchrotron Elettra and the laser at free electron Fermi. Besides this, it study new technology and technique to improve the capabilities and work activities. A lot of the latest improvements are based on robots, that also after the covid and latest with the exploding diffusion of LLMs, allow an interaction of the users of the light sources to communicates and interact with the robots in the facility to obtain information and services as transport of packages.

¹<https://www.elettra.eu/>

Chapter 2

Research questions and objectives

Chapter 3

Scope and limitations